INTERNATIONAL CYBERSECURITY AND
DIGITAL FORENSICS ACADEMY

GRC 103 Practical Laboratory

Advanced Cyber security Risk Quantification and Control Investment Analysis

BY

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SECUREBANK FINANCIAL

BOARD CYBERSECURITY RISK BRIEFING
SecureMobile Banking Platform
Reporting Period: October 2026
Prepared for: Board of Directors

1. Executive Summary

A cybersecurity assessment of SecureBank Financial's SecureMobile banking platform identified three significant vulnerabilities affecting transaction processing, customer data protection, and account security. The estimated annualized financial exposure is **$35.0 million**, with Database Injection accounting for approximately 89% of the total.

The proposed security control program requires an initial investment of **$700,000**, with annual maintenance costs of **$105,000**. Based on the current assessment assumptions, the program could reduce estimated annual risk exposure by approximately **$27.0 million (77.04%)**, leaving projected residual exposure of approximately **$8.04 million**.

Management attention is required to prioritize remediation of Database Injection, strengthen API authentication and session security, and ensure that control effectiveness is independently validated.

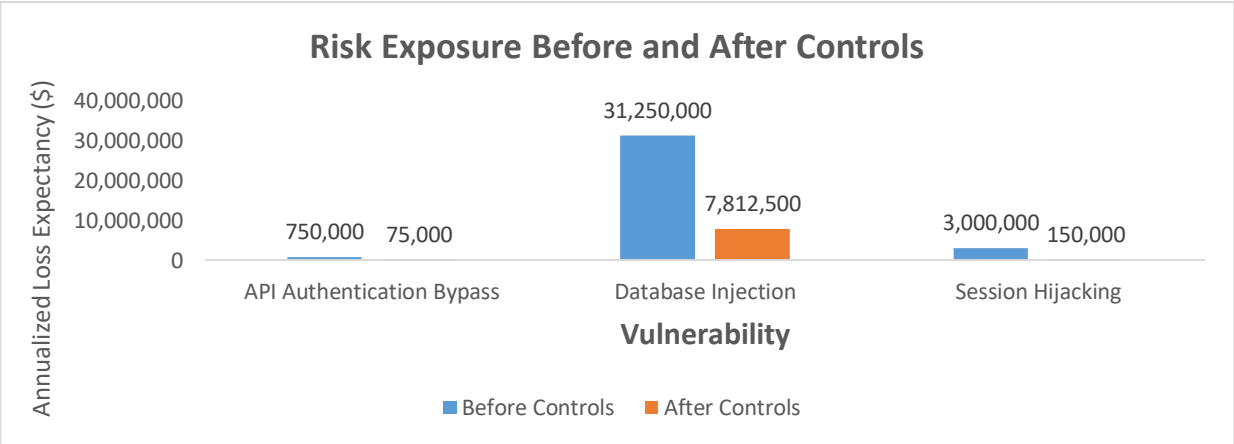


Fig 1: Risk Exposure Before and After Controls

2. Principal Cyber security Risks

Risk	Estimated Annual Exposure	Board-Level Implication
Database Injection	$31.25 million	Potential exposure of customer personal and financial information, with associated regulatory and notification costs
Session Hijacking	$3.00 million	Potential account takeover, unauthorized transactions, and customer losses
API Authentication Bypass	$750,000	Potential unauthorized fund transfers and compromise of transaction integrity
Total	**$35.00 million**	**Material exposure requiring coordinated remediation**

Database Injection is the primary financial concern. API Authentication Bypass also warrants prompt attention given the potential for a single incident to result in a loss of up to $5 million.

3. Recommended Investment

The proposed control program comprises three security investments.

Recommended Control	Initial Investment	Annual Maintenance
Advanced API Security Gateway	$350,000	$50,000
Web Application Firewall (WAF)	$150,000	$25,000
Multi-Factor Authentication Enhancement	$200,000	$30,000
Total	**$700,000**	**$105,000**

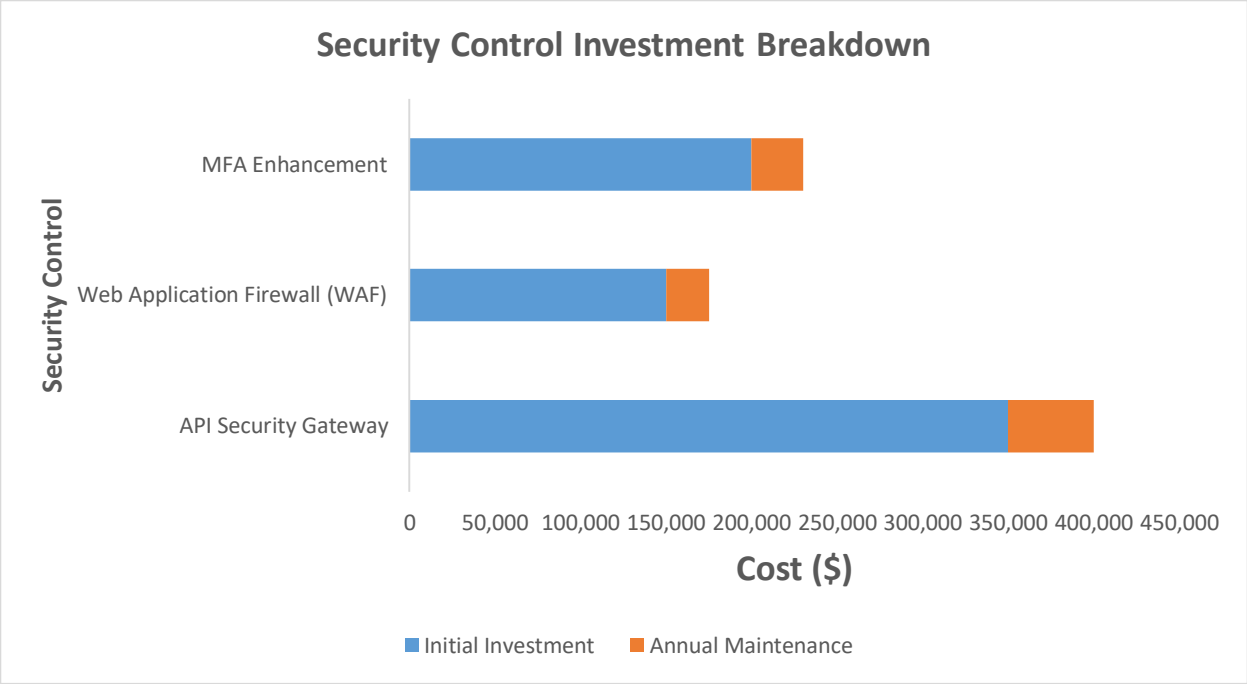


Fig 2: Security Control Investment Breakdown

Fig 2: Compare initial investment and annual maintenance costs for each proposed control.

The proposed program should be supported by direct remediation of identified vulnerabilities. In particular, the Database Injection finding requires corrective action at the application and database level; a WAF should be treated as an additional protective layer rather than a substitute for fixing the underlying weakness.

4. Expected Risk Reduction and Financial Outlook

Measure	Estimated Outcome
Current annual risk exposure	$35.00 million
Expected annual risk reduction	$26.96 million
Projected residual annual exposure	$8.04 million
Expected reduction in exposure	77.04%
Initial investment	$700,000
Annual maintenance	$105,000
Total first-year cost	$805,000

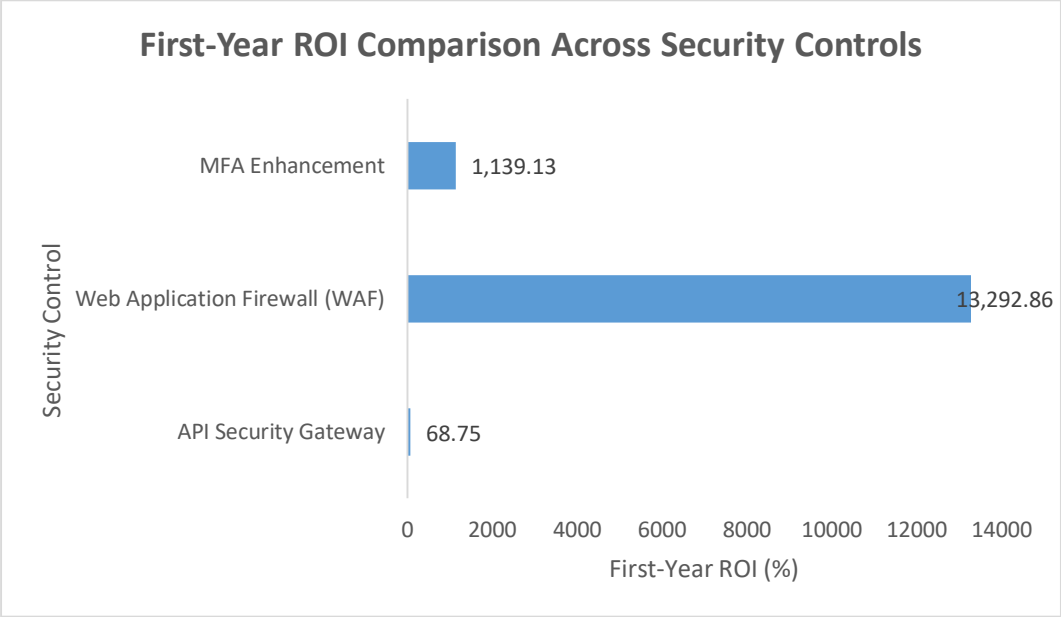


Fig 3: First-Year ROI Comparison

Fig 3: Compare the estimated first-year ROI of the API Security Gateway, WAF, and MFA Enhancement.

The estimated returns are based on assumed control effectiveness and the mapping of each control to a specific vulnerability. They should be regarded as indicative projections rather than guaranteed savings. The projected residual exposure also requires reassessment following remediation and control testing.

5. Implementation and Oversight

The proposed program will be delivered in three stages:

- **Immediate — First 30 days:** Validate findings, implement containment measures, and begin direct remediation of the highest-priority vulnerabilities.
- **Short term — Within 90 days:** Deploy and configure the proposed security controls, complete corrective actions, and conduct security retesting.
- **Long term — Within one year:** Monitor key risk indicators, reassess residual exposure, conduct follow-up penetration testing, and report material changes in risk to management.

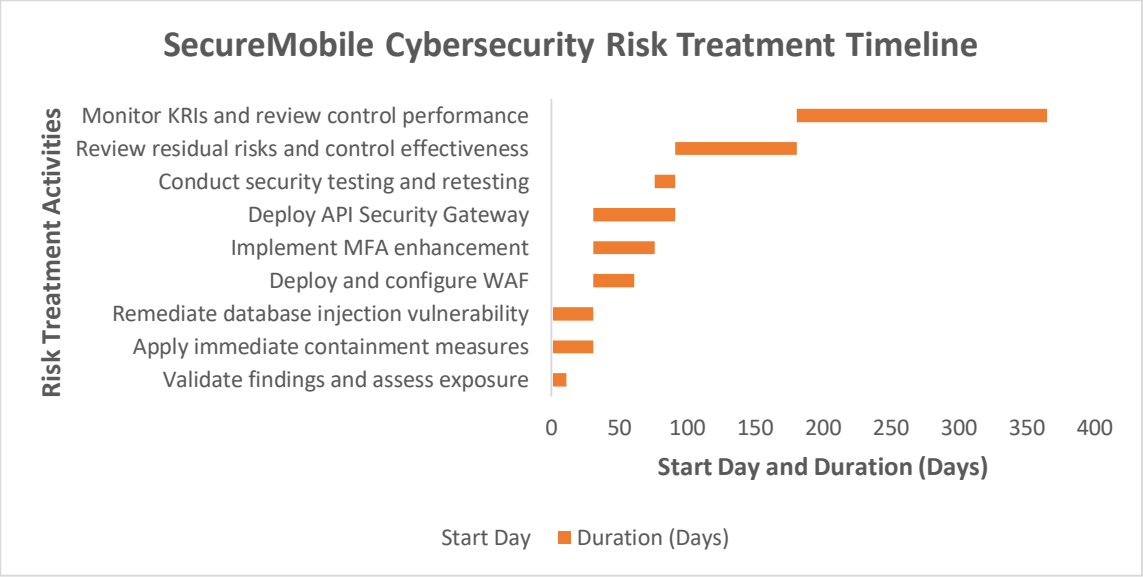


Fig 4: SecureMobile cyber security risk treatment

Fig 4: Show the proposed schedule for immediate actions, short-term controls, follow-up testing, and long-term monitoring.

Progress should be tracked against assigned owners, agreed deadlines, remediation evidence, and control effectiveness measures. Risks remaining above approved tolerance should be escalated to the appropriate risk owner.

6. Board-Level Recommendation

The Board is requested to:

1. **Approve in principle** the proposed $700,000 initial investment and $105,000 annual maintenance budget, subject to management validation of the control design, cost assumptions, and expected benefits.
2. **Prioritize Database Injection remediation** and require evidence that the underlying vulnerability has been corrected.
3. **Require implementation assurance**, including independent security testing and confirmation that the controls operate effectively.
4. **Request periodic risk reporting** on remediation progress, residual exposure, material incidents, and exceptions requiring risk acceptance.

7. Conclusion

SecureMobile has an estimated annual cyber security risk exposure of $35 million, primarily driven by Database Injection. The proposed security program offers a potentially substantial reduction in exposure at a projected first-year cost of $805,000.

Timely remediation, validated control effectiveness, and continued oversight are essential to protect customer information, transaction integrity, regulatory standing, and business continuity.

Board decision required: Approve the proposed investment in principle, subject to validation of the underlying assumptions and a defined implementation and assurance plan.